RAFFLES INSTITUTION
H2 Mathematics 9758
2025 Year 6 Term 3 Revision 2 (Delta)
Topic(s): Complex Numbers and Sequences and Series (Solutions)

Source of Question: 9740/2010/I/Q3

1 The sum, S_n, of the first n terms of a sequence $u_1, u_2, u_3, \ldots$ is given by $S_n = n(2n+c)$, where c is a constant.

(i) Find u_n in terms of c and n. [3]

(ii) Find a recurrence relation of the form $u_{n+1} = \mathrm{f}(u_n)$. [2]

Solution:

(i)

For $n \geq 2$, $u_n = S_n - S_{n-1}$

$$= n(2n+c) - (n-1)(2(n-1)+c)$$
$$= 2n^2 + cn - (2n^2 - 2n + cn - 2n + 2 - c)$$
$$= 4n - 2 + c$$

When $n = 1$, $u_1 = S_1 = 2 + c = 4(1) - 2 + c$

Thus, u_1 follows the form of $u_n = 4n - 2 + c$ when $n = 1$.

Thus, $u_n = 4n - 2 + c$ for $n \geq 1$.

(ii)

$$u_n = 4n - 2 + c$$
$$u_{n+1} = 4(n+1) - 2 + c$$
$$= 4n + 4 - 2 + c$$
$$= 4n - 2 + c + 4$$
$$= u_n + 4$$

Source of Question: 9740/2012/I/Q3(modified)

2 A sequence $u_1, u_2, u_3, \ldots$ is given by

$$u_1 = 2 \text{ and } u_{n+1} = \frac{3u_n - 1}{6} \text{ for } n \geq 1.$$

(i) Find the exact values of u_2 and u_3, and the value of u_{10} to 5 decimal places. [3]

(ii) It is given that $u_n \to l$ as $n \to \infty$. Showing your working, find the exact value of l. [2]

(iii) Describe the behaviour of the sequence. [1]

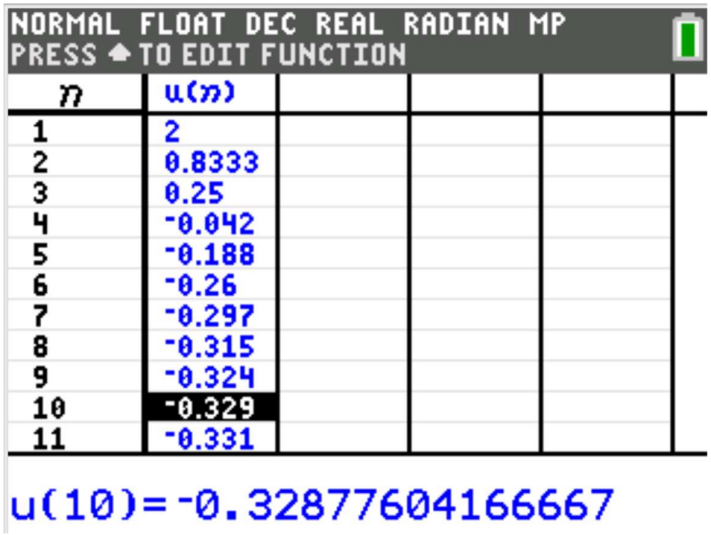
Solution:

(i)

$$u_2 = \frac{3(2)-1}{6} = \frac{5}{6}$$

$$u_3 = \frac{3\left(\frac{5}{6}\right)-1}{6} = \frac{1}{4}$$

From the GC, $u_{10} = -0.32878$ (5 dp)



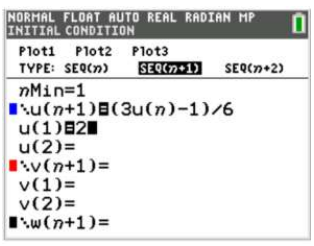
(ii)

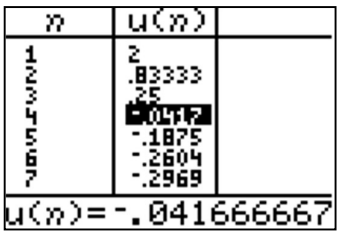
As $n \to \infty$, $u_n \to l$ and $u_{n+1} \to l$

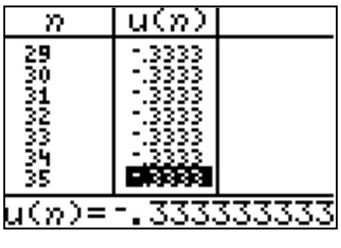
We have $l = \frac{3(l)-1}{6} \Rightarrow 6l - 3l = -1 \Rightarrow l = -\frac{1}{3}$

(iii)

The sequence decreases and converges to $-\frac{1}{3}$.







Source of Question: EJC Prelim 9758/2025/P2/Q2

3 A sequence $u_1, u_2, u_3, \ldots$ is defined by

$$u_{n+1} = 2u_n(1-u_n)$$

for all positive integers n.

(a) Find the possible limits of the sequence, if the sequence is convergent. [2]

(b) For each of the following, evaluate u_2 and u_3, and state whether the sequence is convergent:

- $u_1 = 0$,
- $u_1 = 0.01$,
- $u_1 = -0.01$. [6]

Solution:

(a)	Let L be the limit of the sequence $u_{n+1} = 2u_n(1-u_n)$. $L = 2L(1-L)$ $2L^2 - L = 0$ $L(2L-1) = 0$ $L = 0$ or $L = \frac{1}{2}$
(b)	When $u_1 = 0$, $u_2 = 0$, $u_3 = 0$, sequence is convergent. When $u_1 = 0.01$, $u_2 = 0.0198$ (exact), $u_3 = 0.0388$ (3 s.f.), sequence is convergent. When $u_1 = -0.01$, $u_2 = -0.0202$ (exact), $u_3 = -0.0412$ (3 s.f.), sequence is not convergent.

Source of Question: DHS Prelim 9740/2014/I/Q9(a)

4 A sequence of positive numbers $x_1, x_2, x_3, ..., x_n$ is a geometric progression. Show that the sequence of numbers $y_1, y_2, y_3, ..., y_n$ given by the relation,

$$y_n = \log_k x_n + k, \quad \text{where } k \text{ is a positive constant,}$$

is an arithmetic progression. [4]

Solution:

$y_n - y_{n-1}$
$= \log_k x_n + k - (\log_k x_{n-1} + k)$
$= \log_k x_n - \log_k x_{n-1}$
$= \log_k \frac{x_n}{x_{n-1}}$
$= \log_k r$ (a constant, where r is the common ratio)

Since the difference between any two consecutive terms is a constant, $\{y_n\}$ is an arithmetic sequence.

Source of Question: HCI Prelim 9758/2022/02/Q4

5 (a) (i) Find $\sum_{r=1}^{k}\left[\left(-\frac{1}{2}\right)^{r+1}+\ln(r+1)\right]$ in terms of k. Simplify your answer. [4]

(ii) Hence determine if $\sum_{r=1}^{\infty}\left[\left(-\frac{1}{2}\right)^{r+1}+\ln(r+1)\right]$ exists. [1]

(b) The first term of an arithmetic series is positive. The sum of the first 6 terms of the series is 4.5, and the product of the first four terms of the series is 0. Find the 13th term of the series. [4]

Solution:

(a) (i)

$$\sum_{r=1}^{k}\left[\left(-\frac{1}{2}\right)^{r+1}+\ln(r+1)\right]$$

$$=\sum_{r=1}^{k}\left(-\frac{1}{2}\right)^{r+1}+\sum_{r=1}^{k}\ln(r+1)$$

$$=\sum_{r=1}^{k}\left[\left(-\frac{1}{2}\right)^{r+1}\right]+\ln 2+\ln 3+\ln 4+\ldots+\ln(k+1)$$

$$=\left(\frac{1}{4}\right)\left[\frac{1-\left(-\frac{1}{2}\right)^{k}}{1-\left(-\frac{1}{2}\right)}\right]+\ln\left[(k+1)!\right]$$

$$=\frac{1}{6}\left[1-\left(-\frac{1}{2}\right)^{k}\right]+\ln\left[(k+1)!\right]$$

(a) (ii)

As $k\to\infty$,

$\frac{1}{6}\left[1-\left(-\frac{1}{2}\right)^{k}\right]\to\frac{1}{6}$, $\ln\left[(k+1)!\right]\to\infty$ and hence $\sum_{r=1}^{k}\left[\left(-\frac{1}{2}\right)^{r+1}+\ln(r+1)\right]\to\infty$.

Therefore, the sum to infinity of the series does not exist.

(b)

Let a be the first term of AP and d be the common difference.

$S_6=4.5$

$\Rightarrow\frac{6}{2}(2a+5d)=4.5$

$\Rightarrow 2a+5d=1.5$

$u_1u_2u_3u_4=0$

$\Rightarrow a(a+d)(a+2d)(a+3d)=0$

$\therefore d=-a$ or $-\frac{a}{2}$ or $-\frac{a}{3}$

When $d = -a$,
$\Rightarrow 2a + 5(-a) = 1.5$
$\Rightarrow a = -\frac{1}{2}$ (rej. $\because a > 0$)

When $d = -\frac{a}{2}$,
$\Rightarrow 2a + 5\left(\frac{-a}{2}\right) = 1.5$
$\Rightarrow a = -3$ (rej. $\because a > 0$)

When $d = -\frac{a}{3}$,
$\Rightarrow 2a + 5\left(\frac{-a}{3}\right) = 1.5$
$\Rightarrow a = 4.5$

$\therefore T_{13} = 4.5 + 12(-1.5) = -13.5$

Source of Question: TJC Prelim 9758/2020/02/Q1

6 A sequence of numbers $u_1, u_2, u_3, \ldots$ has a sum S_n where $S_n = \sum_{r=1}^{n} u_r$.

(i) It is known that the sum of the first three terms of the sequence is 17 and $\sum_{n=1}^{3} S_n = 30$.

Furthermore, the third term of the sequence is twice the first term of the sequence. Find the first three terms of the sequence. [3]

(ii) A constant k is subtracted from each term of the sequence so that the first three terms of the modified sequence are now three consecutive terms of a geometric progression with a common ratio greater than 1.
Find the value of k and the common ratio of the geometric progression. [3]

Solution:

(i)

$$u_1 + u_2 + u_3 = 17 \text{--- (1)}$$
$$S_1 + S_2 + S_3 = 30$$
$$u_1 + (u_1 + u_2) + (u_1 + u_2 + u_3) = 30$$
$$3u_1 + 2u_2 + u_3 = 30 \text{ --- (2)}$$
$$u_3 = 2u_1 \Rightarrow 2u_1 - u_3 = 0 \text{ --- (3)}$$

From GC, first 3 terms of sequence are $u_1 = 4$, $u_2 = 5$ and $u_3 = 8$.

(ii)

After subtracting k, the three terms are consecutive terms of a GP and hence

$$\frac{8-k}{5-k} = \frac{5-k}{4-k}$$
$$\Rightarrow 32 - 12k + k^2 = 25 - 10k + k^2$$
$$\Rightarrow k = \frac{7}{2}$$
$$\Rightarrow r = \frac{8-\frac{7}{2}}{5-\frac{7}{2}} = 3$$

Source of Question: TJC Prelim 9758/2019/I/Q6

7 An arithmetic sequence has first term a and common difference d, where a and d are non-zero. The ninth, tenth and thirteenth terms of the arithmetic sequence are the first three terms of a geometric sequence.

(i) Show that $a = -\frac{15}{2}d$. [3]

(ii) The sum of the first n terms of the arithmetic sequence is denoted by S_n. Find the value of S_{16}. [2]

(iii) Given that the kth term of the arithmetic sequence is the fourth term of the geometric sequence, find the value of k. [3]

Solution:

(i)

Since the first three terms of the GP are $a+8d, a+9d, a+12d$, the common ratio is

$$\frac{a+9d}{a+8d} = \frac{a+12d}{a+9d}$$

$$\Rightarrow \quad (a+9d)^2 = (a+8d)(a+12d)$$
$$\Rightarrow \quad a^2+18ad+81d^2 = a^2+20ad+96d^2$$
$$\Rightarrow \quad 2ad = -15d^2$$

$\therefore \quad a = -\frac{15}{2}d$ since d is non-zero

(ii)

$$S_{16} = \frac{16}{2}\left[2a+(16-1)d\right]$$
$$= 8\left[-15d+15d\right] = 0$$

(iii)

Common ratio of GP: $\frac{a+9d}{a+8d} = \frac{-\frac{15}{2}d+9d}{-\frac{15}{2}d+8d} = 3$

k th term of AP = 4th term of GP

$$a+(k-1)d = (a+8d)r^3$$
$$-\frac{15}{2}d+(k-1)d = \left(-\frac{15}{2}d+8d\right)\cdot 3^3$$
$$k-\frac{17}{2} = \frac{27}{2} \quad \text{since } d \neq 0$$
$$k = 22$$

Source of Question: ACJC Prelim 9758/2021/01/Q7

8 Mr Chan is interested in investing in a savings account with an interest rate of 3.2% per year, so that on the last day of each year, the amount in the savings account on that day is increased by 3.2%. He decided to invest \$8 000 on the first day of each year, starting from 2020.

(i) Show that the amount in the account at the end of n years after the interest has been added is given by $\$\,258\,000(1.032^n - 1)$. [3]

After 10 years, Mr Chan stopped investing in this savings account. If he does not withdraw any money from this savings account, the savings account will still continue to generate interest of 1.5% per year on the amount in the account, so that on the last day of each year the amount in the account on that day is increased by 1.5%. Assuming that Mr Chan does not withdraw any money from the savings account,

(ii) by the end of which year will the total interest be first more than \$35 000 from the day that Mr Chan first started his saving account? [3]

At the age of 55, Mr Chan is able to receive a monthly pay-out over a period of 20 years from the savings account if he did not withdraw before. The monthly pay-out in the first year is \$850. The monthly pay-out for each subsequent year is an increment of \$$D$ from the monthly pay-out of the previous year. The total pay-out to Mr Chan at the end of 20 complete years is \$352 200.

(iii) Given that the monthly pay-out is \$1 500 for the m th year of the 20 year period, where m is a positive integer, find m. [3]

Solution:

(i)

nth year	Beginning	End
1	8000	$8000(1.032)$
2	$8000(1.032)+8000$	$8000(1.032)^2+8000(1.032)$
3	$8000(1.032)^2+8000(1.032)$ $+8000$	$8000(1.032)^3+8000(1.032)^2$ $+8000(1.032)$
…		…
n		$8000(1.032)^n+8000(1.032)^{n-1}$ $+\ldots 8000(1.032)^2+8000(1.032)$

(ii) After 10 years, total amount in the account $= 258000\left(1.032^{10} - 1\right)$

$= 95522.18995$

Let k be the additional number of years required.

Total interest earned after $10+k$ years $= 95522.18995(1.015)^k - (8000\times 10)$

$$95522.18995(1.015)^k - (8000\times 10) \geq 35000$$
$$(1.015)^k \geq 1.2039$$
$$k \geq \frac{\ln 1.2039}{\ln 1.015} \quad \Rightarrow \quad k \geq 12.5$$
$$\text{least } k = 13$$
$$2020+10+13-1 = 2042$$

Hence, he will have a total interest be first more than \$35 000 at the end of 2042.

(iii)

Year	Pay-out per month	Pay-out per year
1	850	12(850)
2	850+D	12(850+D)
3	850+2D	12(850+2D)
⋮	⋮	

$$\text{Total pay-out at the end of 20 years} = 12(850)+12(850+D)+...+12(850+19D)$$
$$= 12(850+(850+D)+...+(850+19D))$$
$$= 12\times\frac{20}{2}\left(2\times 850+(20-1)D\right)$$
$$120\left(1700+(20-1)D\right) = 352200$$
$$19D = 1235$$
$$D = 65$$

To find the year m with a monthly pay-out of \$1500:

$$850+(m-1)(65) = 1500$$
$$(m-1)(65) = 650$$
$$m = 11$$

$\therefore$ The monthly pay-out is \$1500 in the 11th year.

Source of Question: SAJC Prelim 9758/2020/01/Q3

9 Mary and Leo signed up for a 42 km marathon and decided to run every day to train for the marathon.

On the first day of Mary's preparation, she begins her training by running 12 km. For each odd numbered day of training, she increases the distance by 2km each time. For each even numbered day of training, she runs the same distance as the day before. Her daily distance ran is then $\{12,12,14,14,16,16,\cdots\}$.

(i) Show that Mary first runs 42 km on the 31st day of training. [2]

Leo starts his training with a 12 km run on the same day as Mary, but decides to increase the distance every day by $x\%$ such that he would be running 42 km on the same day as Mary first runs 42 km.

(ii) Find the value of x, giving your answer correct to 2 decimal places. [2]

(iii) Find the total distance covered by both runners on the day they have completed their first 42 km run at the end of the training. [3]

Solution:

(i)

Let us consider an arithmetic progression with $a = 12$ and $d = 2$, where n is the number of odd days.

Then $u_n = 12 + (n-1)(2)$.

Let $u_n = 42$, then $12 + (n-1)(2) = 42$ gives $n = 16$.

Hence it requires 16 odd numbered days to complete 42 km.

Starting on 1st day, the last day is given by 1 + (16-1)(2) = 31.

Therefore Mary completed 42 km on the 31st day of training. (shown)

(ii)

Let us consider a GP, with $a = 12$, $r = \left(1 + \frac{x}{100}\right)$ and $n = 31$ for Leo to first run 42 km

Using $u_n = ar^{n-1}$

$$12\left(1 + \frac{x}{100}\right)^{31-1} = 42$$

using GC, $x = 4.2643 = 4.26$ (2 d.p.)

(iii)

Total distance covered by Mary
$= 2 \times$ distance covered during 1st 15 odd numbered days
+ distance covered on 31st day

$$\text{Total distance covered} = \left(2\left[\frac{15}{2}\left(2(12) + (15-1)\times 2\right)\right] + 42\right)$$

$$+ \frac{12\left(1 - (1+0.042643)^{31}\right)}{1 - (1+0.042643)}$$

$$= 822 + 745.516$$

$$= 1567.516$$

$$= 1570 \ (3\text{s.f.})$$

Source of Question: VJC Prelim 9758/2022/01/Q5

10 Do not use a calculator in answering this question.

The complex number z satisfies the equation

$$z^2-(4+\mathrm{i})z+2(\mathrm{i}-t)=0,$$

where t is a real number. It is given that one root is of the form $k-k\mathrm{i}$, where k is real and positive. Find t and k, and the other root of the equation. [7]

Solution:

$(k-k\mathrm{i})^2-(4+\mathrm{i})(k-k\mathrm{i})+2(\mathrm{i}-t)=0$

$\left(k^2-2k(k\mathrm{i})+(k\mathrm{i})^2\right)-\left(4k+k\mathrm{i}-4k\mathrm{i}-k\mathrm{i}^2\right)+2(\mathrm{i}-t)=0$

$\left(k^2-2k^2\mathrm{i}-k^2\right)-\left(4k-3k\mathrm{i}+k\right)+2(\mathrm{i}-t)=0$

$\left(-5k-2t\right)+\mathrm{i}\left(2+3k-2k^2\right)=0$

Compare

Real part: $-5k-2t=0$ $\quad---(1)$

Imaginary parts: $2+3k-2k^2=0$ $\quad---(2)$

From (2): $-2k^2+3k+2=0$

$(2k+1)(k-2)=0$

$k=-\frac{1}{2}$ or $k=2$

$\left(\text{reject } k=-\frac{1}{2} \text{ since } k \text{ is real and positive}\right)$

Substitute $k=2$ into (1), $t=-5$

$z^2-(4+\mathrm{i})z+2(\mathrm{i}+5)=0$

$z=\frac{(4+\mathrm{i})\pm\sqrt{(4+\mathrm{i})^2-4(1)(2\mathrm{i}+10)}}{2}$

$=\frac{(4+\mathrm{i})\pm\sqrt{15+8\mathrm{i}-(8\mathrm{i}+40)}}{2}$

$=\frac{(4+\mathrm{i})\pm\sqrt{-25}}{2}$

$=\frac{4+\mathrm{i}\pm5\mathrm{i}}{2}$

$=2+3\mathrm{i}$ or $2-2\mathrm{i}$

Hence the other root is $2+3\mathrm{i}$

Source of Question: NJC Prelim 9758/2022/01/Q4(i)

11 Two complex numbers w and u satisfy the equations

$$w^* - 2\mathrm{i}u = 8 \quad \text{and} \quad (2\mathrm{i}-1)w + 2u^* = 4.$$

Find w and u, giving your answers in the form $x + \mathrm{i}y$, where x and y are real. [4]

Hence find $|u^* - w|$ and $\arg(u^* - w)$. [2]

Solution:

Method 1

$$w^* - 2\mathrm{i}u = 8 \quad \Rightarrow \quad w^* = 8 + 2\mathrm{i}u \quad \Rightarrow \quad w = 8 - 2\mathrm{i}u^* \quad \cdots (1)$$

$$(2\mathrm{i}-1)w + 2u^* = 4 \quad \cdots (2)$$

Put (1) in (2):

$$(2\mathrm{i}-1)(8 - 2\mathrm{i}u^*) + 2u^* = 4$$

$$16\mathrm{i} - 8 + 4u^* + 2\mathrm{i}u^* + 2u^* = 4$$

$$(6 + 2\mathrm{i})u^* = 12 - 16\mathrm{i}$$

$$u^* = \frac{12 - 16\mathrm{i}}{6 + 2\mathrm{i}} = 1 - 3\mathrm{i} \quad \text{from GC}$$

$$u = 1 + 3\mathrm{i}$$

From (1), $w = 8 - 2\mathrm{i}u^* = 8 - 2\mathrm{i}(1 - 3\mathrm{i}) = 2 - 2\mathrm{i}$

Method 2

$$w^* - 2\mathrm{i}u = 8 \quad \Rightarrow \quad \mathrm{i}w^* + 2u = 8\mathrm{i} \quad \Rightarrow \quad -\mathrm{i}w + 2u^* = -8\mathrm{i} \quad \cdots (1)$$

$$(2\mathrm{i}-1)w + 2u^* = 4 \quad \cdots (2)$$

(2) – (1):

$$(3\mathrm{i}-1)w = 4 + 8\mathrm{i}$$

$$w = \frac{4 + 8\mathrm{i}}{3\mathrm{i} - 1} = 2 - 2\mathrm{i} \quad \text{from GC}$$

From $\mathrm{i}w^* + 2u = 8\mathrm{i}$, $u = \dfrac{8\mathrm{i} - \mathrm{i}w^*}{2}$

$$= \frac{8\mathrm{i} - \mathrm{i}(2 + 2\mathrm{i})}{2} = 1 + 3\mathrm{i}$$

$|u^* - w| = |1 - 3\mathrm{i} - 2 + 2\mathrm{i}| = |-1 - \mathrm{i}| = \sqrt{2}$

$\arg(u^* - w) = \arg(-1 - \mathrm{i}) = -\frac{3\pi}{4}$

Source of Question: JJC JC2 Mid-Year CT 9758/2018/01/Q4(iii)(iv)

12 Given $u = 1+\sqrt{3}\mathrm{i}$ and $v = 1-\mathrm{i}$ are roots of the equation $z^4 + az^3 + bz^2 + cz + d = 0$, where a, b, c and $d \in \mathbb{R}$,

(a) find the values of a, b, c and d. [4]

The complex numbers $\mathrm{i}u$ and v^* are represented by the points P and Q respectively on an Argand diagram.

(b) Find the area of triangle OPQ. [3]

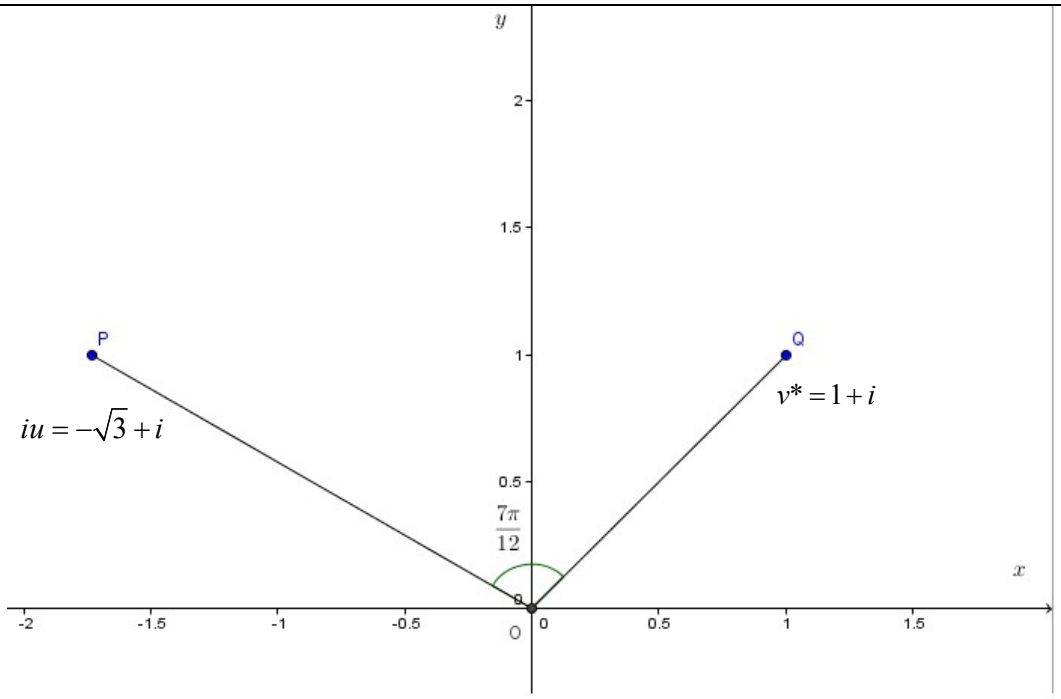
Solution:

(a)

Since $a, b, c, d \in \mathbb{R}$, $u^* = 1-\sqrt{3}\mathrm{i}$ and $v^* = 1+i$ are also roots of the equation $z^4 + az^3 + bz^2 + cz + d = 0$.

$$\Rightarrow \left(z-\left(1+\sqrt{3}\mathrm{i}\right)\right)\left(z-\left(1-\sqrt{3}\mathrm{i}\right)\right)\left(z-(1-\mathrm{i})\right)\left(z-(1+\mathrm{i})\right)=0$$
$$\Rightarrow \left(z^2-2z+4\right)\left(z^2-2z+2\right)=0$$
$$\Rightarrow z^4-4z^3+10z^2-12z+8=0$$
$$\therefore a=-4,\ b=10,\ c=-12,\ d=8$$

(b)



Point P: $iu = i(1+\sqrt{3}i) = -\sqrt{3}+i$

Point Q: $v^* = 1+i$

	Length of OP $= \lvert iu \rvert = \lvert -\sqrt{3} + i \rvert = \sqrt{\left(-\sqrt{3}\right)^2 + 1^2} = 2$ Length of OQ $= \lvert v^* \rvert = \lvert 1 + i \rvert = \sqrt{1^2 + 1^2} = \sqrt{2}$ Argument of i$u = \frac{\pi}{3} + \frac{\pi}{2} = \frac{5\pi}{6}$ Argument of $v^* = \frac{\pi}{4}$ $\Rightarrow \angle POQ = \frac{5\pi}{6} - \frac{\pi}{4} = \frac{7\pi}{12}$ Area of triangle $POQ = \frac{1}{2} \times 2 \times \sqrt{2} \times \sin\left(\frac{7\pi}{12}\right)$ $= 1.37$ units^2

Source of Question: ASRJC Prelim 9758/2025/P2/Q6

13 The points P and Q are represented by the complex numbers p and q respectively, where

$$p = \sqrt{2} + \sqrt{2}\text{i}\ ,\ \arg(q) = \frac{2}{3}\pi \text{ and } \lvert q \rvert = 2\,.$$

(a) Find $\lvert p \rvert$ and $\arg(p)$ in exact form. [2]

(b) Sketch the points P and Q on an Argand diagram. [2]

(c) Use the Argand diagram, or otherwise, to find Re(q) and Im(q), giving your answers in exact form. [2]

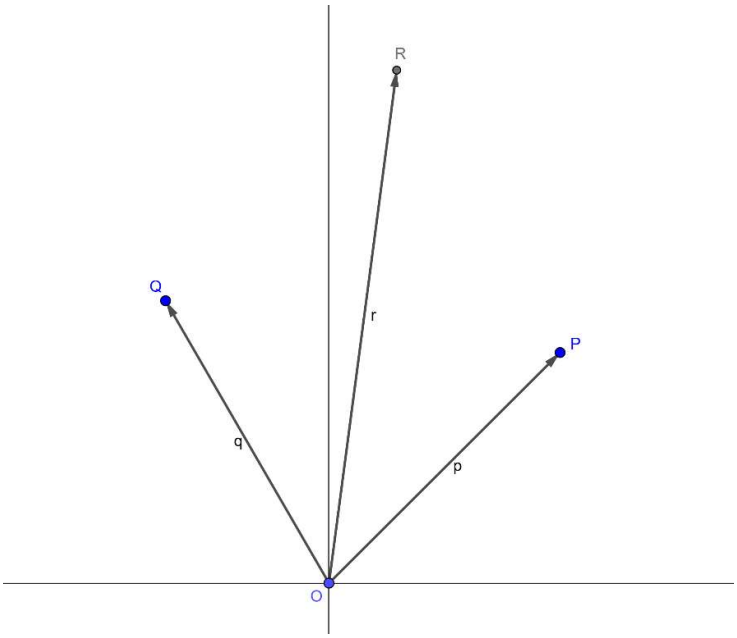
(d) The point R is represented by the complex number $p+q$. What can you deduce about the shape of quadrilateral $OPRQ$, where O is the origin? [1]

(e) By considering $\arg(p+q)$ or otherwise, show that $\tan\left(\frac{11}{24}\pi\right) = \sqrt{6} + \sqrt{3} + \sqrt{2} + 2$. [3]

Solution:

(a)	(a) $\lvert p \rvert = \sqrt{\left(\sqrt{2}\right)^2 + \left(\sqrt{2}\right)^2} = 2$ $\arg(p) = \tan^{-1}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{\pi}{4}$

(b)



(c) Let F be the foot of perpendicular from Q to the x-axis.

$$\mathrm{Re}(q) = -OF = -|q|\cos\frac{\pi}{3} = -2\times\frac{1}{2} = -1 \qquad \text{or} \qquad \mathrm{Re}(q) = |q|\cos\frac{2\pi}{3} = -1$$

$$\mathrm{Im}(q) = FQ = |q|\sin\frac{\pi}{3} = 2\times\frac{\sqrt{3}}{2} = \sqrt{3} \qquad\qquad \mathrm{Im}(q) = |q|\sin\frac{2\pi}{3} = \sqrt{3}$$

(d) Quadrilateral $OPRQ$ is a rhombus.

(e)

$$\text{First, } \arg(p+q) = \frac{1}{2}\left(\frac{\pi}{4}+\frac{2\pi}{3}\right)$$

$$= \frac{11\pi}{24}$$

$$\text{Also, } p+q = \sqrt{2}+\sqrt{2}\mathrm{i}-1+\sqrt{3}\mathrm{i}$$

$$= \left(\sqrt{2}-1\right)+\left(\sqrt{3}+\sqrt{2}\right)\mathrm{i}$$

$$\text{Therefore, } \tan\left(\frac{11}{24}\pi\right) = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}-1}$$

$$= \frac{\sqrt{3}+\sqrt{2}}{\sqrt{2}-1}\times\frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$= \sqrt{6}+\sqrt{3}+\sqrt{2}+2$$

Source of Question: YIJC Prelim 9758/2025/P2/Q2 (modified)

14 Do not use a calculator in answering this question.

It is given that $z = \sqrt{3} - \mathrm{i}$ and $w = 2\mathrm{i}z$.

(a) Find the modulus and argument of z. [2]

O, A, B and C represent the complex numbers 0, z, $z+w$ and w respectively.

(b) Illustrate these four points on a single diagram. [2]

(c) Deduce the modulus and argument of w. [2]

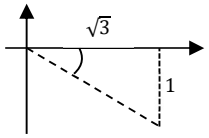
(d) Identify the shape of the quadrilateral $OABC$ and state its area. [2]

Solution:

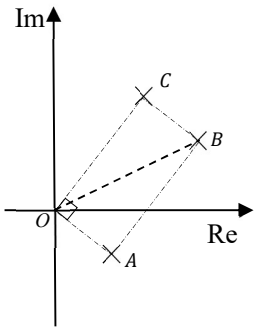
(a)

$$|z| = \sqrt{\left(\sqrt{3}\right)^2 + 1^2} = 2$$

$$\arg(z) = -\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$$



(b)



(c) Using the geometrical relationship between z & $\mathrm{i}z$, and referring to the diagram,

$$|w| = 2|\mathrm{i}z| = 2|z| = 2 \times 2 = 4$$

$$\arg(w) = \arg(z) + \frac{\pi}{2} = -\frac{\pi}{6} + \frac{\pi}{2} = \frac{\pi}{3}$$

(d) Since O, A, B, C forms a parallelogram and $\angle AOC = \frac{\pi}{2}$, thus $OABC$ is a **rectangle**.

Area of rectangle = length × breadth = $2 \times 4 = 8$.

Source of Question: NJC Prelim 9758/2025/P1/Q2

15 Do not use a calculator in answering this question.

(i) Find the values of z and w that satisfy the equations $(1+\mathrm{i})z+2w=-2+4\mathrm{i}$ and $3z-w=4+2\mathrm{i}$, expressing your answers in the form $c+d\mathrm{i}$, where $c,d\in\mathbb{R}$. [4]

(ii) Points W and Z represent w and z found in part **(i)**. Find $\dfrac{w}{z}$ in the form $p+\mathrm{i}q$, where $p,q\in\mathbb{R}$. Hence, state the transformation that maps line segment OZ onto line segment OW. [2]

Solution:

(i)

$(1+\mathrm{i})z+2w=-2+4\mathrm{i}$

$3z-w=4+2\mathrm{i} \quad\Leftrightarrow\quad 6z-2w=8+4\mathrm{i}$

Solving simultaneously by elimination:

$$(6+1+\mathrm{i})z=-2+4\mathrm{i}+8+4\mathrm{i}$$

$$(7+\mathrm{i})z=6+8\mathrm{i}$$

$$z=\frac{6+8\mathrm{i}}{7+\mathrm{i}}\times\frac{7-\mathrm{i}}{7-\mathrm{i}}$$

$$=\frac{42-6\mathrm{i}+56\mathrm{i}+8}{50}$$

$$=\frac{50+50\mathrm{i}}{50}=1+\mathrm{i}$$

$$w=3z-4-2\mathrm{i}$$

$$=3(1+\mathrm{i})-4-2\mathrm{i}=-1+\mathrm{i}$$

(ii)

$$\frac{w}{z}$$

$$=\frac{-1+\mathrm{i}}{1+\mathrm{i}}$$

$$=\frac{-1+\mathrm{i}}{1+\mathrm{i}}\times\frac{1-\mathrm{i}}{1-\mathrm{i}}$$

$$=\frac{-1+\mathrm{i}+\mathrm{i}+1}{1^2+1^2}$$

$$=\mathrm{i}$$

$w=\mathrm{i}z$

Alternative

$$\frac{w}{z}$$
$$= \frac{-1+\mathrm{i}}{1+\mathrm{i}}$$
$$= \frac{\mathrm{i}^2+\mathrm{i}}{1+\mathrm{i}}$$
$$= \frac{\mathrm{i}(\mathrm{i}+1)}{1+\mathrm{i}}$$
$$= \mathrm{i}$$

$$w = \mathrm{i}z$$

Rotating line segment OZ by $\frac{\pi}{2}$ anti-clockwise about the origin will give OW.

Or

Line segment OW is an anti-clockwise rotation about the origin by $\frac{\pi}{2}$ from line segment OZ.